# Swastik Mishra

sgm9198@nyu.edu | (929) 326-0418 | New York, NY | swastikmishra.com

## EDUCATION

**New York University** **June 2025 - May 2027**
M.S. in Financial Engineering | GPA: 3.93/4 *New York City, United States*
Coursework: Derivative Securities, Algorithmic Trading and High Frequency Finance, Quantitative Trading, Hedge Fund Strategies

**Bennett University** **Sept 2021 - June 2025**
B.Tech. in Computer Science & Engineering (Specialization: Data Science) | GPA: 3.72/4 *Greater Noida, India*
Coursework: Probability & Statistics, Time Series Analysis, Machine Learning, Data Structure & Algorithms, High Performance Computing

## SKILLS & RECOGNITION

**Technical Skills:** Python (pandas, NumPy, statsmodels, SciPy, scikit-learn, PyTorch), C++, SQL, R
**Competitions:** 1st Tech @ NYU Buildathon (FinTech Track, 2026), Cornell Trading Competition 2025 (Team NYU)
**Awards:** Chanakya Fellowship - ₹100k research grant (IIT Indore, 2023), Spotlight Award (Publicis Media Division, 2023)
**Certifications:** Bloomberg Market Concepts, Simulation & Modeling of Natural Processes (Université de Genève)

## WORK EXPERIENCE

**Publicis Groupe** **Aug. 2023 - Dec. 2024**
Data Science Intern, Solutions Team, Media Division (South Asia) *Gurugram, India*
- Applied hierarchical constrained optimization at brand and portfolio levels, improving mROI by ~1.5% across APAC markets
- Built latent factor structural SEM across 50+ touchpoints & 4 stages for a multi-market SE Asia study, identifying brand equity drivers
- Engineered Python-R integration & Streamlit/AWS interface for Markriti (Publicis MMM platform on Meta's Robyn) adopted globally
- Delivered MMM solutions for Fortune 500 brands, developing tools & dashboard products generating $240k+ revenue annually
- Built a LangChain agentic platform integrating 7+ live and proprietary data sources across 15+ markets, driving $300k+ in revenue

**Indian Institute of Science, Education and Research** **May 2023 - July 2023**
Summer Research Intern, Weather and Climate Modeling Lab *Mohali, India*
- Optimized features in Himalayan rainfall prediction by evaluating climate indicators, improving stacked GRU $R^2$ ~30 ppts (~40→70)
- Built multivariate time series pipeline on 145 years of climate data (1871-2016) with 60-month lookback & 40-year held-out test split

## PROJECTS

### ★ ACADEMIC

**NYU - Vertically Integrable Project - Forecasting Energy Markets (under Prof. Amine Aboussallah)** **Jan. 2026 - Ongoing**
- Building statistical & ML models for next-day WTI-oil price direction forecasting with past prices & 31 FRED-MD macro features

**Cross-Sectional Equity Momentum on Historical S&P 500 Constituents** **Feb. 2026 - Mar. 2026**
- Built survivorship-aware cross-sectional S&P data using historical index membership & daily adjusted returns (1995-2026, Bloomberg)
- Constructed long-short quantile-based rebalanced portfolios; tested signal variants notably momentum, vol-scaling, & regime overlays
- Mapped 31 years of M&A activity to trace corporate evolution and strategic pathways across firms creating systematic delisting dataset

**Arbitrage under Mean Reversion (Boguslavsky & Boguslavskaya, 2004)** **Dec. 2025**
- Replicated optimal framework for OU spreads under power utility, re-deriving HJB, closed-form optimal control & value function
- Deployed live Streamlit dashboard backtesting OU strategy on 30+ real spreads; calibrated parameters via discrete-time regression

### ★ HACKATHONS

**AutoAlphaPods (Cornell AI Hackathon'26)** **Mar. 2026**
- Engineered natural language backtesting pipeline: users input a thesis, Sphinx agent runs backtests, returning strategy metrics and plots
- Built multi-agent alpha research platform where Sphinx DS agents ideate, face adversarial PM critique & backtest if thesis is sound
- Deterministic backtest pipeline ingests live yfinance/Fed data via Supabase; approved strategies regressed against SPY to test for alpha

**FalseMarkets (1st Tech @ NYU Buildathon, FinTech Track)** **Feb. 2026**
- Designed genetic algorithm framework evolving SMA trading rules (fast/slow MA, risk aversion, asset selection) for 25+ crypto assets
- Developed integrated finance chat platform using CoinGecko, yfinance, GNews, & Alpaca for interactive portfolio & market analytics

**SnapIndex (Qualcomm Edge AI Hackathon)** **Sep. 2025**
- Built Windows-native local semantic search (35+ file-types) delivering sub-second directory searches optimized for Snapdragon NPUs